

In T 966/18, which includes detailed technical reasoning, the board was convinced that the medical use of claim 1 was plausible on the basis of several cited documents illustrating the knowledge of the skilled person, taken together with the relevant disclosure of the patent. It concluded that the body of knowledge evidenced had made the skilled person aware of a link between the reduction of α -synuclein aggregation and the treatment of Lewy body disease. Later published evidence was also taken into account and confirmed this conclusion.

On post-published documents, see also chapter II.C.6.8. and the referral decision in T 116/18 (OJ 2022, A76), which, albeit concerned more with inventive step (Art. 56 EPC), also addresses sufficiency of disclosure (Art. 83 EPC) and identifies the following main lines of case law: "ab initio plausibility" (plausibility was ultimately not established in the decisions listed by T 116/18); "ab initio implausibility" (plausibility was ultimately established); and "no plausibility" line of case law. Case pending under G 2/21.

- c) Consideration of post-published documents limited: cannot remedy a fundamental insufficiency of disclosure

Post-published evidence may be taken into account, but only to back-up the findings in the application in relation to the use of the compound(s) as a pharmaceutical (T 609/02, T 950/13).

If the description in a patent specification provides no more than a vague indication of a possible medical use for a chemical compound yet to be identified, more detailed evidence cannot be used later to remedy the fundamental insufficiency of disclosure of such subject-matter (T 609/02). Where the therapeutic effect is a functional technical feature of the claim, the application must disclose the suitability of the product to be manufactured for the claimed therapeutic application and evidence filed later cannot be used to remedy a fundamental insufficiency of disclosure (point established in T 609/02, as summarised in T 1045/13).

Referring to T 609/02, the board in T 433/05 recalled that where a therapeutic application was claimed in the Swiss-type form, attaining the claimed therapeutic effect was a functional technical feature of the claim. As a consequence, under Art. 83 EPC 1973 the application had to disclose the suitability of the product to be manufactured for the claimed therapeutic application (see also T 1685/10). Note, however, that according to G 2/08 (OJ 2010, 456), where the subject matter of a claim is rendered novel only by a new therapeutic use of a medicament, such claim may no longer have the format of a so-called Swiss-type claim as instituted by decision G 5/83 (OJ 1985, 64). Following T 609/02, the board in T 801/06 recalled that a claimed therapeutic effect **may be proven by any kind of data** as long as they clearly and unambiguously reflect the therapeutic effect. Thus, the fact per se that the experiments in the patent were not carried out with a "real" metastasis was not sufficient to deny sufficiency of disclosure.

In T 2571/12 (treatment of schizophrenia with glutathione), the board stated that there was no evidence at all either in the patent or in the available prior art for a therapeutic effect of glutathione precursors for any of the claimed disorders, and hence the post-published

document, allegedly supporting such an effect *inter alia* for bipolar disorders, likewise could not be taken into account for the assessment of sufficiency of disclosure.

In T.1045/13, concerning a second medical use claim worded in accordance with Art. 54(5) EPC, the board stated that the application as filed did not provide any indication of the principle underlying the relationship between the activity of the pharmaceutically active agent, i.e. the NGF, and the therapeutic effect, i.e. the alleviation of the symptoms of the psychological conditions claimed. The description did not provide any information on the mechanism of action of NGF. No background references were cited that linked NGF to the therapeutic effects to be obtained, and no *in vitro* assays were provided to illustrate any such effect. In the complete absence of such information, the experimental evidence on file was of decisive importance. But the evidence in the application as filed did not constitute a sufficient disclosure. Indeed the experimental evidence consisted of eleven examples relating to various conditions but did not cover all the conditions defined in claim 1; each example related to a single patient only, contrary to the common practice in order to allow statistical analysis of the results. Also, in the absence of a control group, placebo effects could not be excluded. The post-published evidence might be taken into account, but only to back up the findings in the patent application. Post-published evidence could not establish sufficiency of disclosure on its own. They did not need therefore to be discussed. For sufficiency of disclosure, it is not relevant what the respondent (patent proprietor) was aware of, but decided not to disclose.

d) Whether opponents can cite post-published documents

See, in this chapter II.C.7.3, the decision in T.1872/16, in which the board stated that opponents were free to use whatever evidence they chose to substantiate their serious doubts. This also applied to the date on which the evidence was generated.

e) Types of evidence of a therapeutic effect and consideration of clinical trials

In T.2218/16 (gene therapy of motor neuron disorders – scAAV9 vector), the board summarised that, according to the established case law, the provision of evidence in the patent application for a claimed effect was not a prerequisite for patentability, if, based on the data in the patent application/patent, or on common general knowledge, it was plausible that a product (here: scAAV9) was suitable for the claimed therapeutic applications (the decision also contains a detailed summary of the case law on the burden of proof; see the full point 32 of the Reasons).

All kinds of experimental data have been accepted by the boards. It has also been repeatedly emphasised that "it is not always necessary that results of applying the claimed composition in clinical trials, or at least to animals are reported" (T.1273/09 citing T.609/02).

In T.421/14 (dosage regime involving twice-daily treatment – multiple sclerosis), the claims related to a further medical use. All objections raised by the appellants (opponents) concerned the credibility of the alleged therapeutic efficacy. The respondent relied on the data and analysis of a clinical trial. It was known that only a proportion of patients

responded to the treatment. The existence of a population of non-responders was also confirmed by the inventors' own results. In the board's view, the existence of these non-responders was not a reason to find the disclosure insufficient, and the treatment of non-responders did not have to be excluded or disclaimed; that a substantial proportion of patients were non-responders was a common phenomenon.

T 338/10 concerned a claim for a second therapeutic use in which the active ingredient was a "first allergen" and the therapeutic use the treatment or prevention of an allergy caused by a different, second allergen. The board found that there was no experimental data in the patent proving that a first allergen could be used to treat an allergy caused by a different allergen.

In **T 1777/12** claims 1 and 10 were medical use claims and related to the use of a PYY agonist in the manufacture of a medicament for treating a metabolic disorder in an obese or overweight subject. Clinical trials are not required to establish suitability. It may suffice that in vitro or in vivo data directly and unambiguously reflect the therapeutic effect on which the claimed therapeutic application relies or, alternatively, that there is an established relationship between the physiological activities of the compound under consideration and the disease in question. The board concluded that the disclosure in the patent demonstrated the suitability of PYY for achieving a beneficial effect in all of the claimed therapeutic applications by reducing weight or weight gain.

A post-published document reporting that no proven effective vaccine against HSV was available did not prove the non-workability of the invented vaccine, since there might have been other reasons, e.g. regulatory reasons, for not producing vaccines according to the invention. Furthermore, for compliance with the requirements of Art. 83 EPC 1973, it was not necessary to undertake and disclose clinical trials (**T 1023/02**).

For the acceptance of sufficient disclosure of a therapeutic application, it is not always necessary for results of clinical trials to be provided at the relevant date, but the patent/patent application must provide some information showing that the claimed compound has a direct effect on a metabolic mechanism specifically involved in the disease. Once this evidence is available from the patent/patent application, post-published evidence may be taken into account to support the disclosure in the patent application (**T 433/05**). In **T 1045/13** the board agreed with the statements made in **T 433/05** and **T 801/06**, but in case **T 1868/16** (suitability of everolimus for treating PNETs), no data or plausible technical concept was present in the patent.

7.2.3 Pharmaceutical compositions

In **T 1616/09** the board pointed out that, for the purposes of Art. 83 EPC, the level of disclosure in the application which is required for claims directed to pharmaceutical compositions or kits is not the same as that which is required for medical-use claims. For claims directed to pharmaceutical compositions or kits it is in principle sufficient that the application provides information which allows the skilled person to produce the composition or kit, and that there are no substantiated doubts that it could indeed be used in therapy. For second-medical-use claims, on the other hand, it is required not only that

the composition itself is disclosed in an enabling way but also that its suitability for the claimed treatment is **plausibly** disclosed in the application. In the case of a claim directed to a pharmaceutical composition comprising two classes of compounds which had both already been used in therapy in the prior art, there was a priori no reason to doubt that such a pharmaceutical composition could be produced; no specific functional effect had to be demonstrated. In the case of second-medical-use claims, if the claimed therapeutic effect was already known to the skilled person at the priority date, it was not necessary to demonstrate it in the application. According to T.1616/09, T.609/02 does not apply to compositions but only to second-medical-use-claims (see also T.1592/12, which also states that it is not sufficient to show that the skilled person can apply the claimed dosage regime, points 16-17 of the Reasons).

7.2.4 Field of pharmaceutical combinations (drug-drug interactions)

See T.391/18 above on the need in the field of pharmaceutical combinations to assess drug-drug interactions for each drug combination.

7.2.5 Medicament directed to a specific group of patients

T.1491/14 concerned a medicament whose novel feature was that it was directed to a specific group of patients. A lack of sufficiency was alleged on the basis, in particular, that the skilled person was not able to identify the group of patients. Contrary to the opponent's arguments, it was not necessary to measure a physical or chemical parameter to identify the patients. This view was confirmed by the expert declarations (standard practice for a physician to question the patient). The fact that there were no standard questionnaires for this purpose did not create an impasse (anamnesis). While there might sometimes be uncertainty, this amounted to a lack of clarity rather than a lack of sufficiency. As the technical features in question (definition of the patient group) were already present in the claims as granted, and the cause of uncertainty was not the amendment, this issue was not within the scope of the opposition appeal proceedings (see G.3/14).

7.3. Level of disclosure required for antibodies

In T.431/96 the skilled person seeking to reproduce the invention would have had to produce monoclonal antibodies by routine methods and test them singly in an assay. Although this might possibly involve some tedious and time-consuming work, it was nothing out of the ordinary since the techniques for the production and selection of hybridomas were common routine techniques at the priority date of the patent in suit.

The board found that the essential issue to be considered in T.601/05 of 2 December 2009 was whether or not the patent enabled the production of human monoclonal antibodies binding with high affinity to soluble TNF and, consequently, whether or not the skilled person could practise the invention over the whole scope of the claim (following T.792/00). On the evidence before the board it did not.

In T.1466/05 the question arose whether the availability of a hybridoma producing one specific antibody together with a general description of the epitope recognised by this

antibody put the skilled person in the position to obtain further antibodies with the same specificity. The board observed that similar questions had arisen in various cases decided by the boards of appeal, and different boards had given different answers depending on the circumstances of each case (T 510/94, T 513/94, T 349/91, T 716/01).

In T 1466/05 the claim was not restricted to monoclonal antibodies defined by reference to the deposited hybridoma. As the application did not disclose any specific antigen for preparing further antibodies as claimed, the board considered that a skilled person seeking to prepare such antibodies would have had to embark on **a research programme** without any teaching in the application as how to achieve the desired specificity which amounted to an undue burden (cited in this respect by T 760/12).

Concerning the second medical use claims (claim 6 in the "Swiss-type" format, claim 7 in the purpose-restricted product claim format) in T 760/12, the technical effect, which was the therapeutic effect, was expressed in the claim. When the technical effect is expressed in the claim, the issue of whether this effect is indeed achieved over the whole scope of the claim is a question of sufficiency of disclosure (G 1/03, OJ 2004, 413, point 2.5.2 of the Reasons). Hence, under Art. 83 EPC, unless this is already known to the skilled person at the priority date, the application must disclose the suitability of the product to be manufactured for the claimed therapeutic application (T 609/02, point 9 of the Reasons). The board concluded that it was not sufficiently disclosed in the patent that a single monoclonal antibody as defined in the claim potentially exerted the therapeutic effect as claimed.

The claim at issue in T 405/06 was directed to immunoglobulins with certain stated features. The question to be answered was whether a skilled person would have found at the filing date in the application as filed a sufficiently clear and complete disclosure of the precise structure of such an immunoglobulin in order to be in a position to prepare it over the broad range of the claim. Although the claim was not limited to immunoglobulins obtained from camelids, the experimental part of the description as a whole and the corresponding figures dealt exclusively with camel immunoglobulins and the general part of the description did not contain a complete disclosure of any non-camelid immunoglobulin either. The requirements of Art. 83 EPC 1973 were thus not satisfied, as the skilled person would be left with the task and burden of finding out how the teaching relating to camelid immunoglobulins could be extended to products of different origins (e.g. human immunoglobulins) falling within the broad area of the claim.

The application the subject of T 433/07 concerned broadly reactive opsonic antibodies that react with common staphylococcal antigens. The board held that the invention was insufficiently disclosed; the application did not disclose either any serotype cross reactive monoclonal antibody or the isolation of an antigen associated with the serotype cross protective response required by the claim. A European patent application containing a claim referring to a method of production had to provide the skilled person with the means to produce the desired product. If this was not the case, this shortcoming could not be overcome by telling him exactly how the desired product had to look and which screening criteria had to be applied to find it.